



IRIDIUM CONSTELLATION

ECE 5610 Course Project

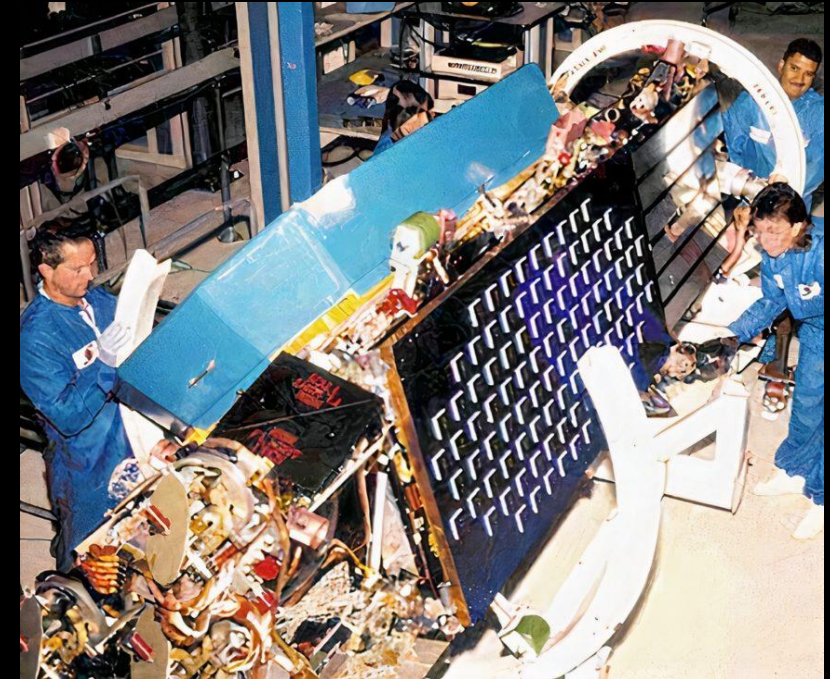
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OVERVIEW

- Background
- Concept of Operations
- User Terminals
- Constellation Design & Orbital Parameters
- Major Subsystems
- Primary Payload – Communications System
- Inter-Satellite Links
- Link Budget Analysis
- Iridium Through History

BACKGROUND

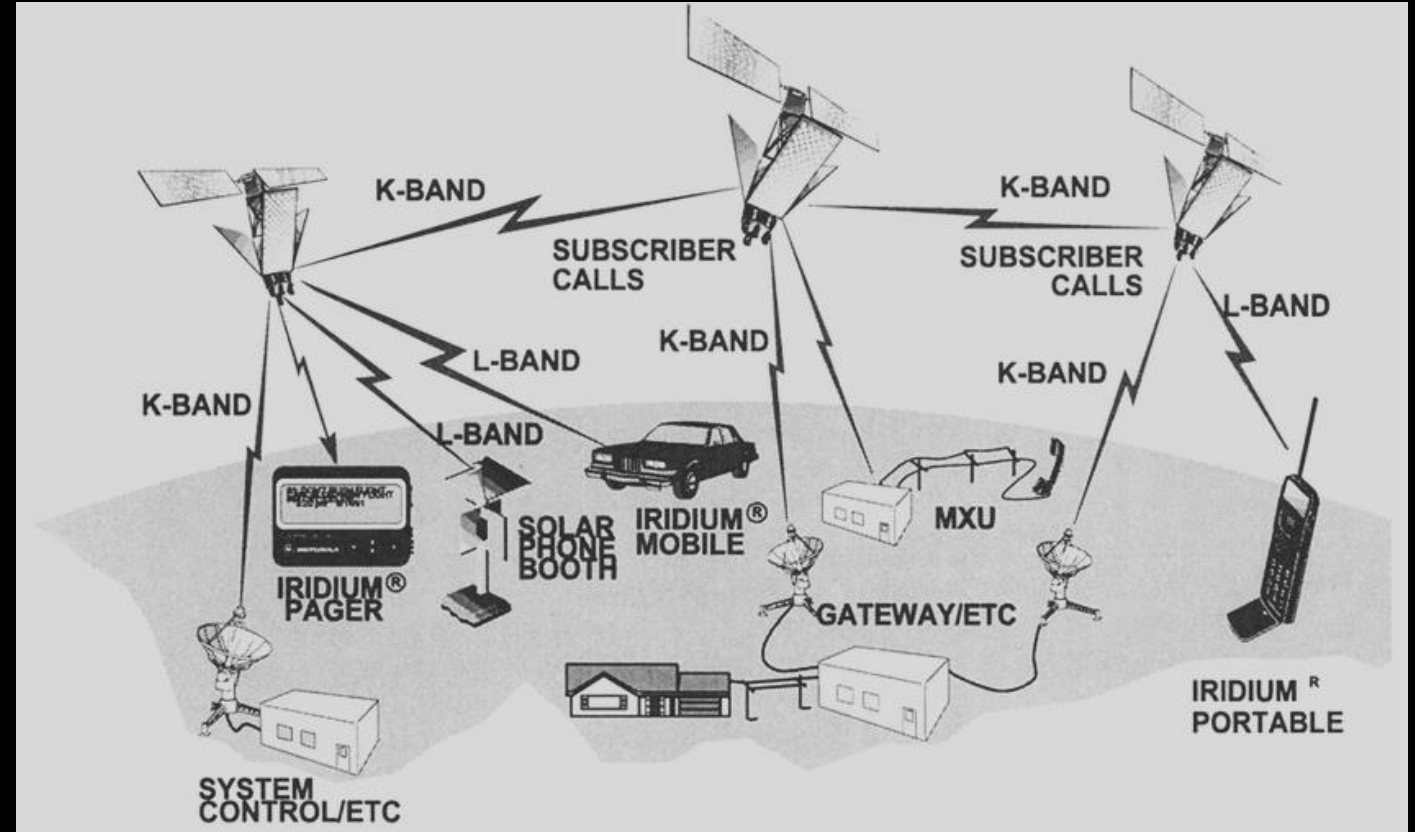
- Conceived by Motorola engineers in 1987 and commercialized through Iridium LLC as a handheld global satellite telephone system in LEO
- Targeted users needing communication beyond terrestrial networks: remote land, oceans, polar regions, aviation, and government use
- Designed around handheld/mobile terminals rather than fixed satellite dishes
- Used a 66-satellite crosslinked mesh to route calls/data globally before reaching a gateway



[Motorola Engineers working on first generation Iridium Satellite](#)

CONCEPT OF OPERATIONS

- The Iridium satellite constellation uses Inter-Satellite links to relay traffic information and downlink to nearest gateways
- The ground user is assigned a satellite and beam by the gateways depending on the best calculated link margin for the user
- Up to 172,000 simultaneous users of Network capacity!



USER TERMINALS

- Low power – Ex. 0.6 W average, 7 W peak for traditional voice unit
- Low gain – 0 to 3 dBi depending on unit & option for external antenna
- Many third-party manufacturers for different types of units



Iridium subscriber unit in use at the South Pole



Iridium Extreme
(Voice, Text)



Iridium 9555
(Voice, Text)



Iridium Solar
(Short-Burst Data)



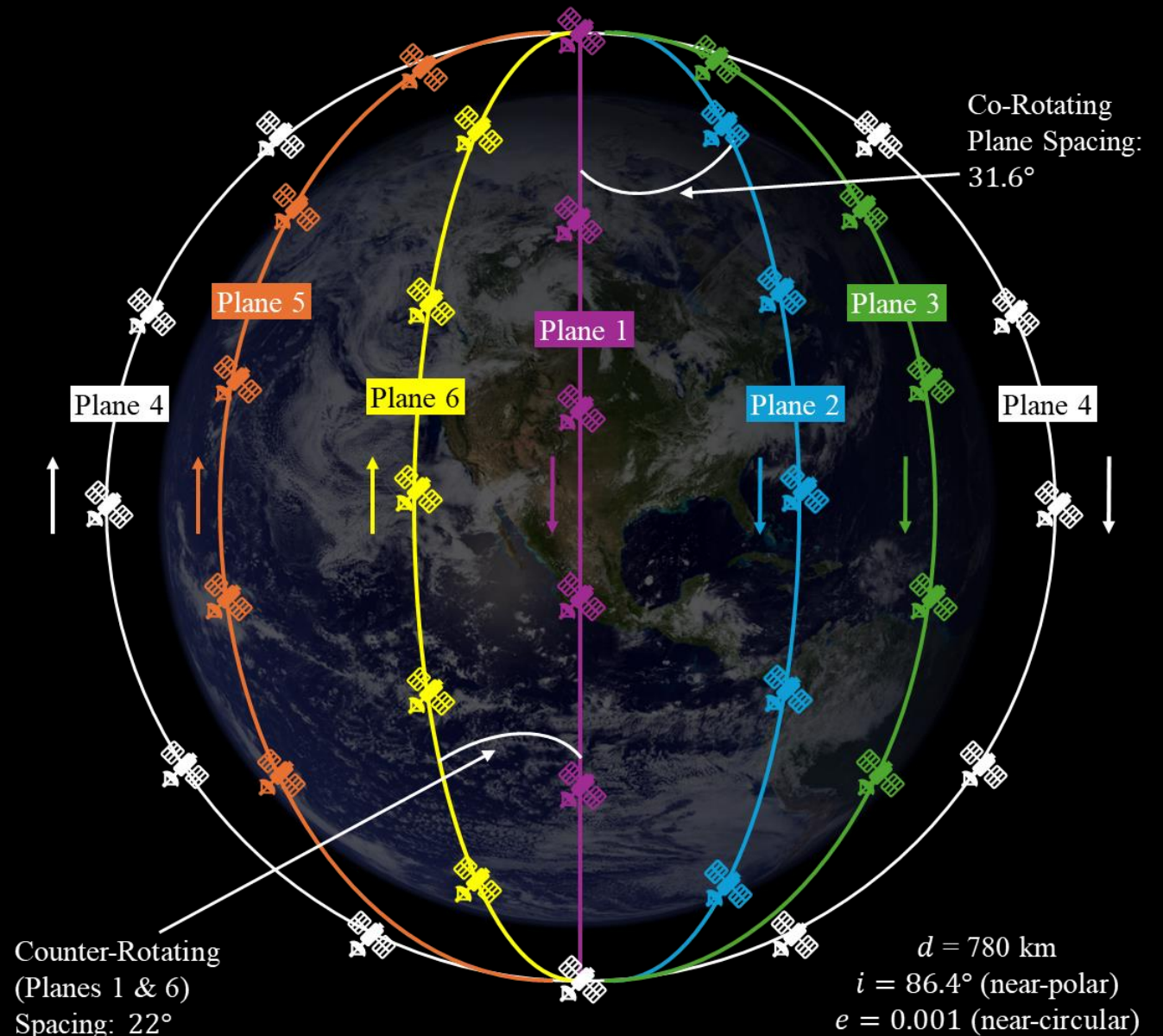
Iridium GO!
(Interface with Other Devices)



Garmin In-Reach Messenger
(Third-Party Text Device)

CONSTELLATION DESIGN & ORBITAL PARAMETERS

- 66-satellite constellation
 - 95 satellites across 22 launches
 - Several spare satellites launched
- Six planes with 11 satellites per plane
- Complete instantaneous global coverage
- LEO altitude, near-polar inclination



MAJOR SUBSYSTEMS

- **TT&C:** Communication with ground station via K-band antennas
- Satellite Control Facility (SCF) in Landsdowne, VA + stations in Italy, Canada, Iceland, and Alaska

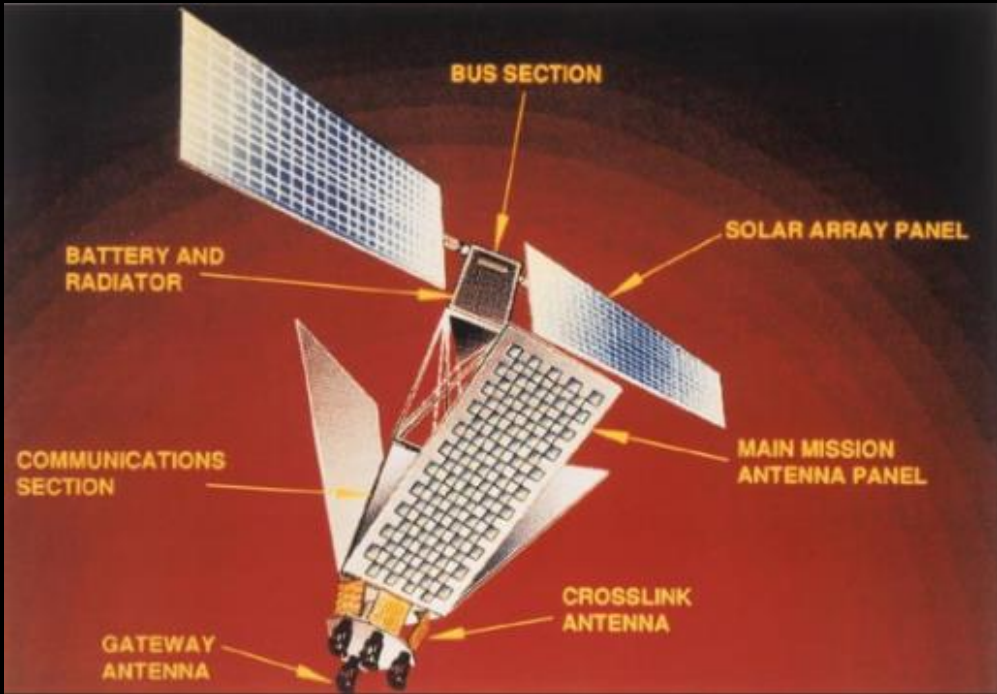


- **EPS:** 2 sun-tracking solar panels (620 W load), 60-Ah battery, & battery radiator assembly – 85% of power goes to comms payload!
- **AOCS:** Three-axis control via momentum bias system, hydrazine propulsion for ascent & station-keeping, and a computer calculating positions of nearby Iridium satellites

PRIMARY PAYLOAD – COMMUNICATIONS SYSTEM

3 systems for each satellite:

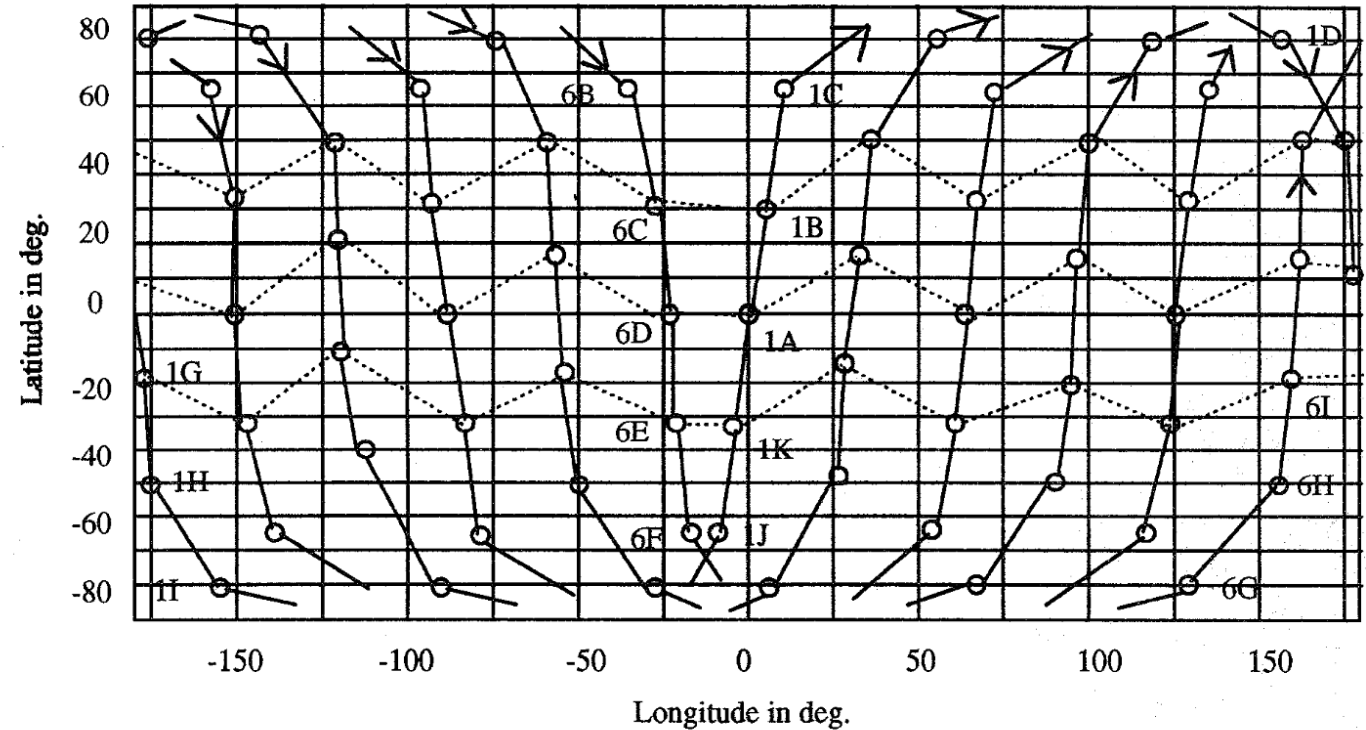
- 1. User-facing Main Mission Antennas (MMAs)
 - 3 Phased array antennas each with 16 beams
 - TDMA & FDMA schemes
 - QPSK user modulation
- 2. TT&C antennas
 - Deployment: omnidirectional narrowband antenna
 - In orbit: omnidirectional wideband antenna
 - Process command and system control data
- 3. Inter-satellite links (ISLs)
 - 4 crosslink antennas (2 fixed, 2 steerable)



	MMAs	TT&C	ISLs
Frequency Band	L-Band	K-Band	K-Band
Frequency Range	1.616-1.6265 GHz	dl:19.4-19.6 GHz	22.55-23.55 GHz
		ul:29.1-29.3 GHz	

INTER-SATELLITE LINKS (ISLS)

- Satellites exchange voice and data traffic information
 - Determine highest link margin for user to determine satellite routing
- 4 ISLs on each satellite
 - 2 Intra-orbit (same orbital plane)
 - 2 Inter-orbit (adjacent orbital planes)
- Uses time-varying routing and antenna pointing angles
- <20% links affected by handover latency longer than 20ms



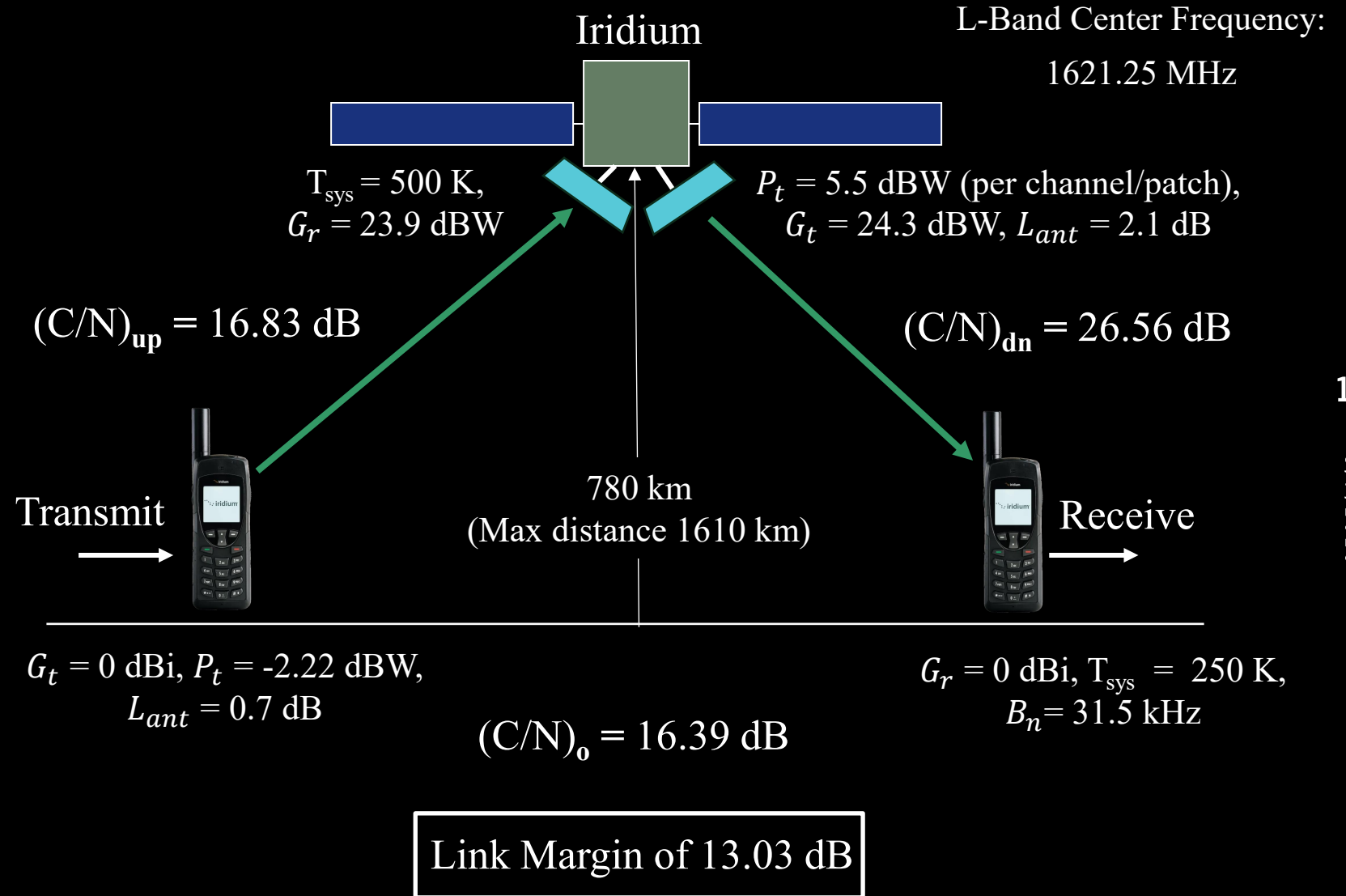
○ Subsatellite points, — Intra-Orbit ISLs, --- Inter-Orbit ISLs

LINK BUDGET ANALYSIS

- Iridium advertises link margin of 16.83 dB
- Minimum uplink received power: $P_r = -148.5$ dBW
- Minimum downlink received power: $P_r = -151.5$ dBW
- Minimum values & noise in the receiver used to calculate minimum C/N values:

$$\left(\frac{C}{N}\right)_{up,req} = 8.13 \text{ dBW}$$

$$\left(\frac{C}{N}\right)_{dn,req} = 5.13 \text{ dBW}$$



IRIDIUM THROUGH HISTORY

Early Commercial Challenges

- 1998: Original system begins service
- 1999: Bankruptcy after high costs, expensive service, and weak adoption

New Funding / New Approach

- 2000: Deorbiting considered, but DoD contract helped keep network alive
- 2001 Onward: Emerged from bankruptcy with niche voice/data focus
 - **Polar Regions:** North Pole expedition (2001) tested Iridium phones under polar conditions
 - **Disaster response:** Used following Hurricane Katrina (2005) and the 2010 earthquake in Haiti when terrestrial networks failed
 - **Long-term markets:** maritime, aviation, government, and remote data links

End of Life / Impact

- 2009: Iridium 33 collision with Kosmos 2251 became a major LEO debris event
- 2019: Services transferred off the legacy fleet to Iridium NEXT; original satellites were deorbited afterward



[Iridium Satellite phones used during Hurricane Katrina](#)



[Iridium OpenPort maritime terminal installed aboard a Spanish commercial vessel](#)

QUESTIONS?

